Full-Array Boolean Logic CIM Macro With Self-Recycling 10T-SRAM Cell for AES Systems

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***Abstract*—Computing in memory (CIM), which alleviates the need to transfer a large amount of data between processor and memory, significantly reducing latency and energy consumption, is a promising new computing architecture for addressing the von Neumann bottleneck problem. This article proposes a CIM array structure composed of self-recycling 10T static random access memory (SRAM) cells, which can realize orthogonal data writing, and multiple Boolean logical operations for the entire array. The self-recycling and full-array activation characteristics are extremely suitable for accelerating diverse data processing algorithms such as the Advanced Encryption Standard (AES). A 4-kb SRAM is implemented in 55-nm CMOS technology to verify the effectiveness of the design. Compared with other state-of-the-art architectures, the throughput and the operating frequency of the proposed CIM macro are increased to 843 GOPS/kb (2.64×) and 823.7 MHz (2.6×), respectively. The energy efficiency reaches 246.9 TOPS/W. When applied to the AES, the energy consumption is 35.77% less than the digital CIM architecture that is not self-recycling.**

***Index Terms*—Full-array activation, in-memory logical operation, self-recycling technology, static random-access memory (SRAM), von Neumann architecture.**

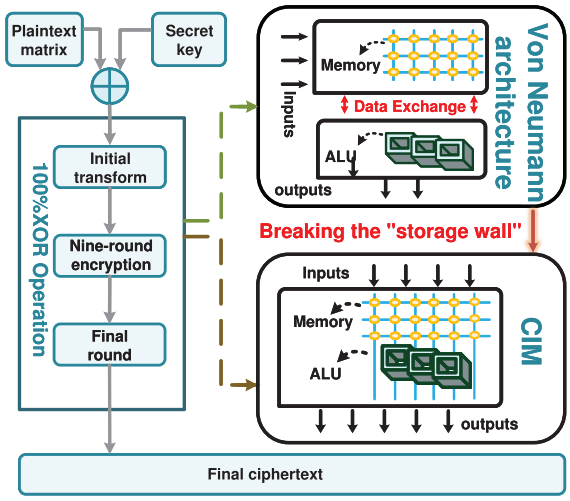


Fig. 1. von Neumann architecture and CIM architecture applied to AES.

I. Introduction

In RECENT years, within the fast-evolving sphere of artificial intelligence (AI), data have become a cornerstone asset, typically incorporating a vast quantity of sensitive and delicate information [1], [2]. The Advanced Encryption Standard (AES), a highly sophisticated and reliable cryptographic technique, is proficient in encrypting data, thus effectively warding off the perils of data theft or malicious alteration during the transmission phase [3]. Its encryption flowchart, as shown in Fig. 1, includes initial transformation, nine rounds of encryption, and a final round of encryption. Each stage involves numerous XOR logical operations, which typically require multiple iterations and frequent memory access. The von Neumann architecture is used to perform logical operations commonly and is implemented by separating memory and compute units. Consequently, data must frequently move between memory banks and computing elements. In addition, as semiconductor technology evolves in accordance with Moore's law, the speed disparity between computing units and storage has been widening progressively, thereby giving rise to the so-called "memory wall" bottleneck.

For the bottleneck of the traditional von Neumann architecture, numerous scholars have put forward the integrated technology of storage and computing [5], [6], [7], [8], [9], [10], [11], [12], [16], [17], [18], [19], [20], [21], [26], [27], [28], [29], among which computing in memory (CIM) is a novel architecture that directly conducts computing within memory, as illustrated in Fig. 1. It breaks through the limitations of the traditional von Neumann system and holds advantages in enhancing throughput, reducing power consumption and latency. Therefore, it has emerged as a prominent solution for augmenting computability.

Static random access memory (SRAM) is used in most of the current memory systems with the advantages of rapid read and write speed, low power consumption, strong reliability, and high scalability [1], [2], [3], [6], [9], [10], [11]. The existing SRAM-based CIM can be classified into mixed-signal and digital-domain CIM. Mixed-signal CIM exhibits significant advantages in power consumption and area

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cost [5], [6], [18], [19], [20], [21]. However, it is susceptible to process, voltage, and temperature (PVT) variations and is more appropriate for fault-tolerant calculations such as applied to the CNN algorithms and transformers. In contrast, digital-domain CIM has high computing accuracy and robustness and, thus, can accelerate AES algorithms [7], [8], [9], [10], [11], [12], [16], [17].

Most of the existing in-memory Boolean logic computing is achieved by opening two rows of cells and then sensing the BL voltage through the sensitive amplifier (SA). In order to store the computing results, in [7], a redundant row based on 8T differential SRAM is needed, which means that additional circuit structure and cycles are required. Hence, it not only increases the area but also leads to higher power consumption for data transmission. In addition, this results in a decrease in computing speed and storage density. Similarly, in [8], [9], and [10], configurable 8T SRAM units, 9T SRAM, and 10T SRAM structures are proposed, respectively, which merely enhance the read-write noise margin and reduce the leakage power consumption within the subthreshold region through the decoupling of read and write paths but does not solve the problem that redundant rows are required for storing the computing results. In addition, they still have to activate only two rows at a time, leading to a rather low throughput consequently.

Therefore, in [11], a CIM based on a 10T-SRAM with bidirectional logical operation is proposed. That CIM can activate the array in parallel with multiple rows or columns for vector logical operations. However, implementing XOR or XNOR operations within the structure necessitates the inclusion of additional NOR gates and SAs. Furthermore, an architecture based on Non-Volatile SRAM (NV-SRAM) technology is proposed in [33], solving the problem of area overhead. The data are overwritten in three phases: 1) state restoration; 2) SRAM read; and 3) RRAM write-back. This process requires computed results to be read from SRAM and then written back to RRAM via peripheral circuits, introducing additional latency and energy overhead. Subsequently, an 8T-SRAM and a high-density dual 4T-SRAM structure are proposed for in-memory logical operation in [12] and [13], respectively. With these structures, the calculation results can be automatically stored back to avoid data migration, thereby improving energy efficiency significantly. Nevertheless, to perform IMP operations that can be applied to intelligent medical device monitoring systems and intelligent vehicle control systems by these two structures, additional circuits are also required to assist them. In view of the above problems such as poor accuracy of mixed-signal CIM, low throughput of Boolean logical operations, extra temporary storage for calculation results, and the need for additional peripheral auxiliary circuits for diverse logical operations, a novel full-array self-recycling 10T-SRAM CIM circuit is proposed in this article. The advantages of the proposed structure are given as follows.

1) *Enhanced Stability and Efficiency:* The self-recycling 10T-SRAM cell offers improved read/write stability.
2) *Functional Diversity:* The self-recycling 10T-SRAM macro supports a wide range of logical operations, including AND, OR, XOR, IMP, and more. This versatility makes the structure highly adaptable to various data processing algorithms, enhancing its practical applicability.
3) *Increased Throughput and Energy Efficiency:* The proposed structure supports two operands orthogonal writing and full-array activation, significantly boosting energy efficiency and throughput, making it well-suited for high-performance computing applications.

The remainder of this article is organized as follows. Section II presents an overview of the proposed self-recycling 10T-SRAM CIM macro, detailing the self-recycling technology integrated into the memory structure. It also discusses the SRAM mode and full-array Boolean logical operations. Section III provides simulation results along with a comprehensive discussion. Finally, the conclusions are summarized in Section IV.

II. Proposed 10T SRAM-Based CIM Macro

This section elaborates, in detail, on the overview of the proposed self-recycling 10T-SRAM CIM macro, the self-recycling technology, the mechanisms of performing data reading and writing operations, and the implementation of full-array Boolean logical operations within the proposed circuits.

A. Overview of the Proposed CIM Macro System

The proposed self-recycling 10T SRAM-CIM macro structure is shown in Fig. 2(a). It mainly includes a self-recycling 10T SRAM array for Boolean operation, a timing control circuit, a mode control circuit, two mutually independent operator input circuits, and a data output circuit.

The 10T-SRAM CIM macro can operate in two modes: the memory mode and the CIM mode. In the memory mode, read and write operations can be performed rowwise or columnwise. Through BLs and BLBs, an operand (operand2) can be written columnwise to or read columnwise from the 64×64 SRAM array, while the other operand (operand1) is written rowwise, jointly controlled by IN1 and IN2, which multiplex the VDD and GND with fewer numbers of signal transitions and transistors used, leading to an improvement in the computing speed and a decrease in power consumption. In the CIM mode, Boolean logical operations can be implemented by controlling the novel self-recycling 10T-SRAM cell. Different from existing works, in this structure, the data in the entire array can be activated and participate in operations simultaneously, and the operation result can be automatically saved back to the original cell (named "self-recycling") when it performs AND, OR, XOR, or IMP operations. Furthermore, to concretely illustrate the self-recycling characteristic, we conduct a comprehensive analysis of the AND logic operation in Section II-D.

B. Self-Recycling 10T-SRAM Cell Design

Compared with the traditional 6T-SRAM cell, the 10T adds transistors ($M3$, $M4$, $M5$, and $M6$) to control the cross-coupled inverters, as shown in Fig. 2(b). The node QB is no longer the drain terminal of $M2$ that is instead isolated from it by $M4$. This may lead to a certain threshold loss when choosing

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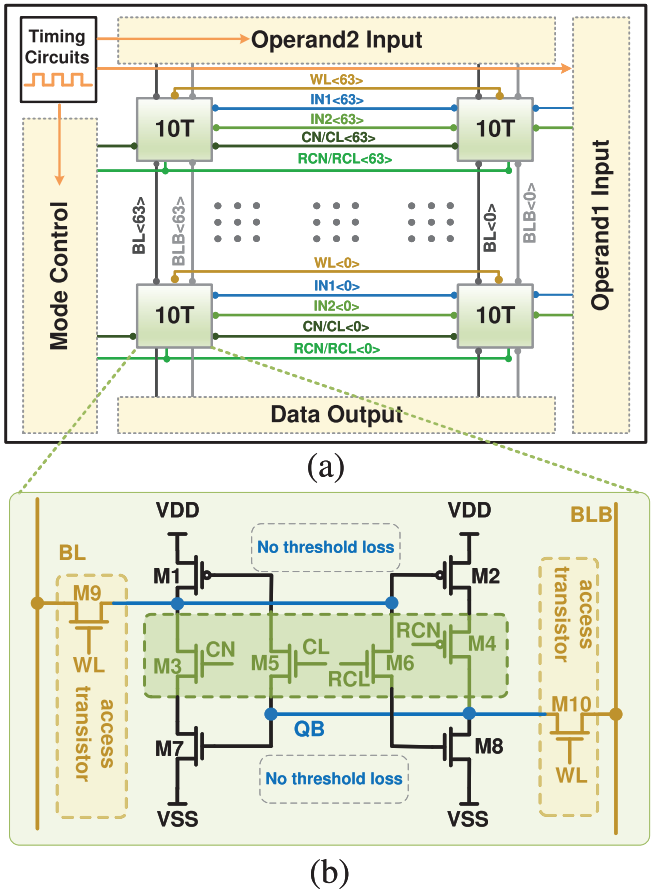


Fig. 2. (a) Overall structure of the proposed CIM macro. (b) Diagram of 10T-SRAM with self-recycling technology.

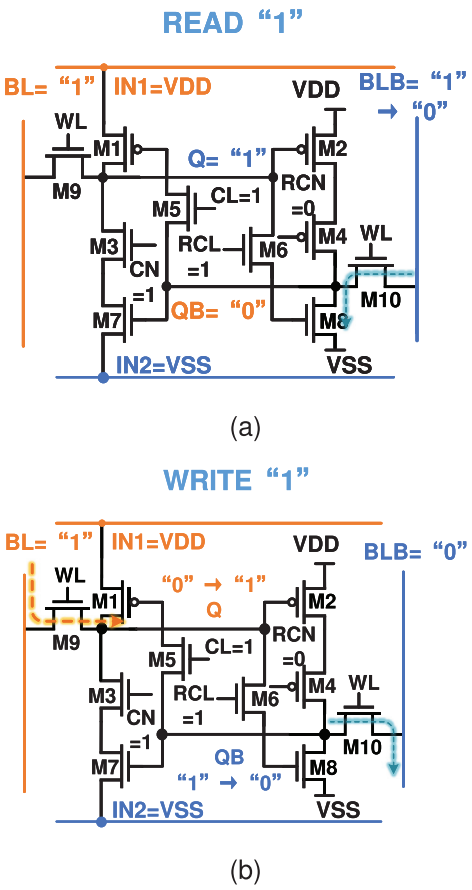


Fig. 3. Reading/writing case of the proposed 10T-SRAM cell: (a) reading "1" and (b) writing "1".

an nMOS transistor as $M4$. For example, when the QB node stores data "1," the voltage may not reach the value of the power supply voltage VDD with a used nMOS transistor. To overcome this problem, a pMOS transistor is used to connect the storage node QB and the pull-up transistor $M2$. Similarly, Q is no longer the drain terminal of $M7$. Therefore, an nMOS transistor is selected for $M3$.

The other two transistors ($M5$ and $M6$) that control the gates of the cross-coupled inverters to govern the process of positive feedback within the cell are selected as nMOS based on the characteristic that the on-resistance of nMOS is smaller than that of pMOS. As a switch, the nMOS has faster speed and better driving ability. Therefore, nMOS is used as the control transistor of the operation to improve the overall performance and response speed.

C. Reading and Writing Operations in Memory Mode

As shown in Fig. 3(a), the operation for reading data "1" is illustrated. During the read operation, the control terminals on the left-hand side are set to CN =1 and CL =1, while, on the right-hand side, RCL =1 and RCN =0. The input terminals are configured as IN1= VDD and IN2= VSS. Both bit lines, BL and BLB, are precharged to a high level "1," and the word line is activated (WL =1). The data stored at the storage node are then read through the two bit lines. Since QB =0, the bitline BLB is discharged through $M8$ and $M10$, creating a voltage difference between BL and BLB. This voltage difference is then used to read out the stored data.

As shown in Fig. 3(b), it is the operation for writing data "1" to Q and data "0" to QB, which will be the operand2 in the CIM mode. For the left-hand side, the operation control terminals are CN = 1 and CL = 1. For the right-hand side, the operation control terminals are RCL = 1 and RCN = 0. The input terminals are IN1 = VDD and IN2 = VSS. BL is set to high level "1," while BLB is set to low level "0," and WL = 1. Assume that the bit Q initially stores "0" and the bit QB initially stores "1." The bitline (BL) charges Q through $M9$. QB discharges to BLB through $M10$, and the data of QB are flipped and written, while the input operand1 can directly input data "1" or "0" through $M1$ and $M7$ in the CIM mode.

D. Boolean Logical Operations in CIM Mode

The self-recycling 10T-SRAM macro enables a variety of full array logical operation functions, such as AND, OR, XOR, and IMP.

1) Boolean and / Nand Operation in CIM Mode: Fig. 4(a) and (c) illustrates the AND/NAND operation process diagram and timing diagrams for the proposed circuit in the CIM mode. In the example shown, the logical operation 1 AND

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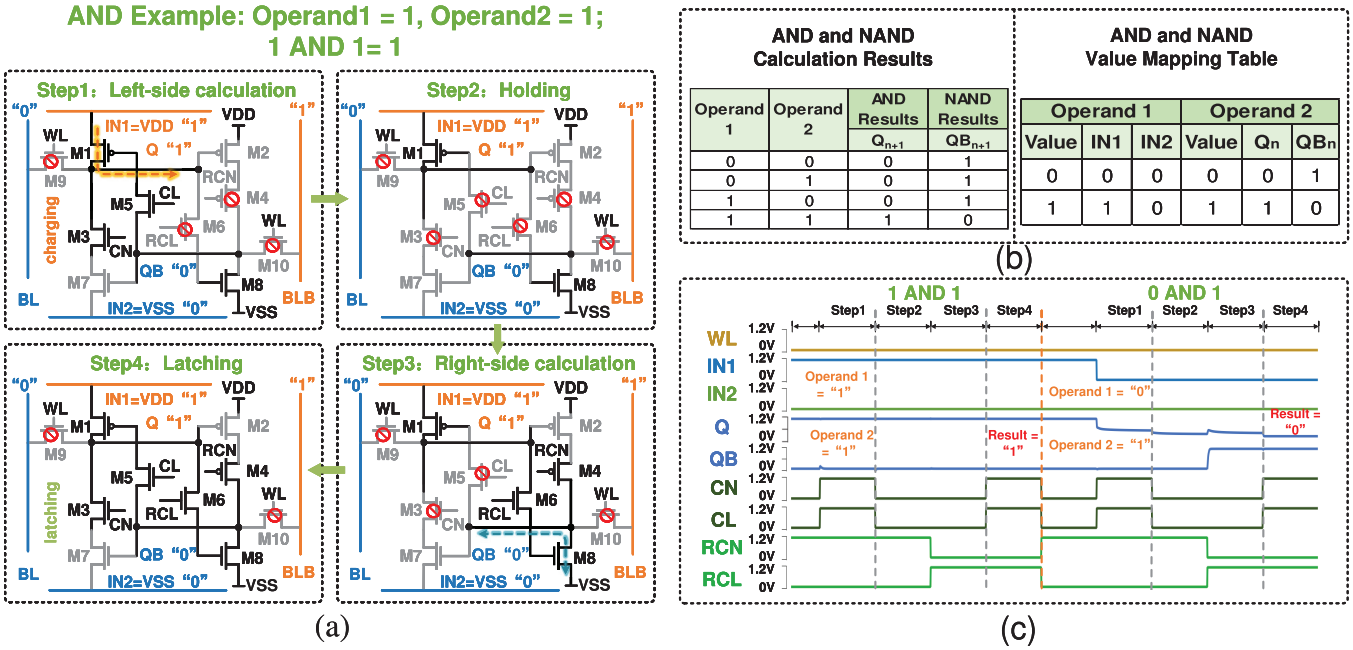


Fig. 4. AND operation of the proposed 10T-SRAM cell: (a) operation procedure, (b) truth and mapping table, and (c) timing diagrams.

1 = 1 is performed. Specifically, the input terminals IN1 and IN2 are configured as 1 and 0, representing input 1, while the storage nodes Q and QB hold 1 and 0. During the Boolean logical operation, WL is set to 0. This ensures that the logical function can be executed without activating the BL or access transistors. By doing so, full array execution is enabled, allowing the circuit to avoid unnecessary energy consumption and minimizing the risk of data corruption, thereby enhancing energy efficiency and data stability. The operation is divided into four steps, detailed as follows.

1) *Left-Half Calculation:* Turn on the transistors $M3$ and $M5$ (CL =1 and CN =1), and turn off the transistors $M4$ and $M6$ (RCL =0 and RCN =1). IN1 charges to Q through transistor $M1$.
2) *Holding:* Turn off transistors $M3$ and $M5$. The values of Q and QB remain 1 and 0.
3) *Right-Half Calculation:* Turn on the transistors $M4$ and $M6$ (RCL =1 and RCN =0). QB discharges to 0 through $M8$.
4) *Latching:* All transistors ($M3$, $M4$, $M5$, and $M6$) are activated, latching the result such that Q and QB reflect the operation outcome.

Fig. 4(b) presents the AND/NAND calculation results along with the corresponding value mapping table. In this table, the input terminal IN2 is fixed at 0. When IN1 equals 0, it represents input operand1 = 0; conversely, when IN1 equals 1, it represents input operand1 = 1. The current states of the storage nodes are used to represent operand2: specifically, Qn = 0 and QBn = 1 indicate operand2 = 0, while Qn = 1 and QBn = 0 correspond to operand2 = 1. The results of the AND/NAND operations are subsequently reflected in the next states, denoted as Qn +1 and QBn +1. For example, if both operand1 and operand2 are 1 (IN1 = 1, IN2 = 0, Qn = 1, and QBn = 0), the next state of Qn +1 and QBn +1 will become 1 and 0, respectively, corresponding to the AND and NAND operation results. By contrast, if operand1 or operand2 is 0, the next state of Qn +1 and QBn +1 will become 0 and 1, respectively. The timing diagrams for the logical operation 0 AND 1 = 0 are shown on the right-hand side of Fig. 4(c).

2) Boolean OR/NOR, XOR / XNOR, and IMP Operation in CIM Mode: The OR/NOR, XOR/XNOR, and IMP operations are similar to the AND/NAND operations, as shown in Fig. 5. The desired logical operations can be achieved by appropriately modifying the coding of IN1 and IN2, which serve as the representations of operand1. For the OR/NOR operation, IN1 must be held at 1, with operand1 represented by IN2, where IN2 = 1 or 0, as illustrated in Fig. 5(b). For the XOR/XNOR operations, the coding is given as follows: when IN1 = 1 and IN2 = 0, it represents input operand1 = 0, whereas, when IN1 = 0 and IN2 = 1, it represents input operand1 = 1, as depicted in Fig. 5(d). When implementing the IMP operation, IN2 is fixed at 1, and the inverse state of IN1 is used to represent operand1, as shown in Fig. 5(f). Upon completion of these operations, the calculated results are reflected in the next states of the storage nodes Qn +1 and QBn +1.

III. Simulation Results

To verify the function of the proposed work, a 4-kb 64×64 CIM macro is implemented and simulated in CMOS 55-nm technology.

A. Stability of the 10T-SRAM Cell

The stability evaluation of the self-recycling 10T-SRAM cell is primarily divided into two key aspects: 1) assessing the stability of the storage node during CIM mode operation and 2) analyzing the read, write, and hold noise margins of

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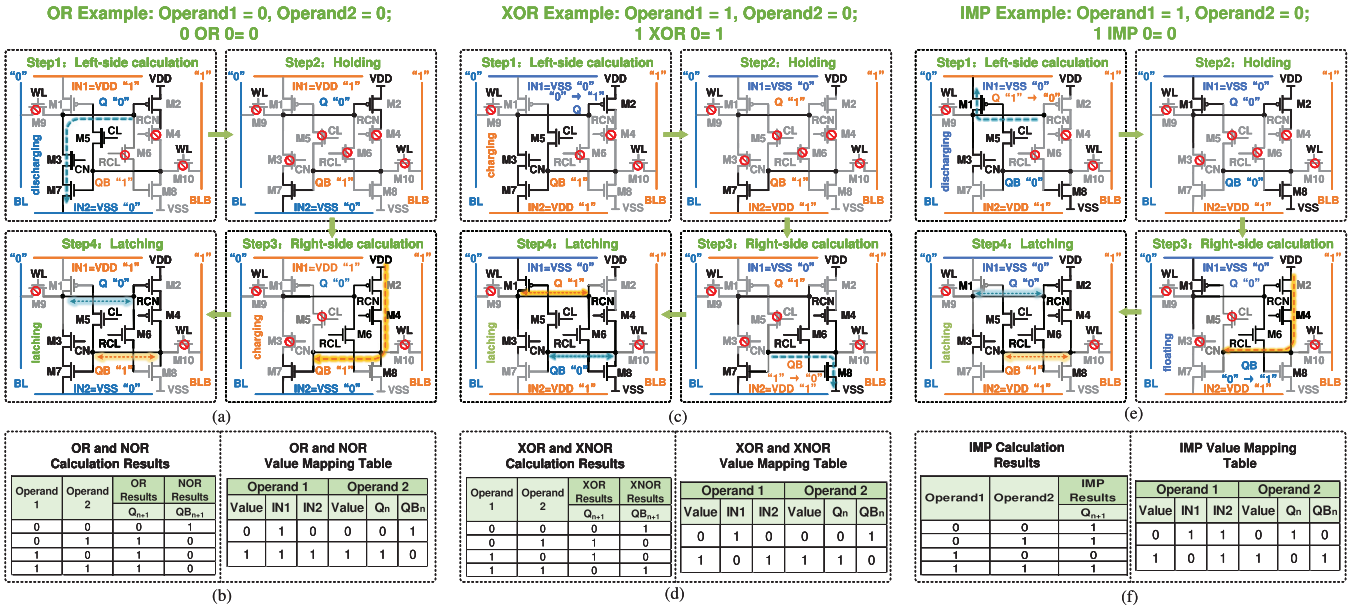
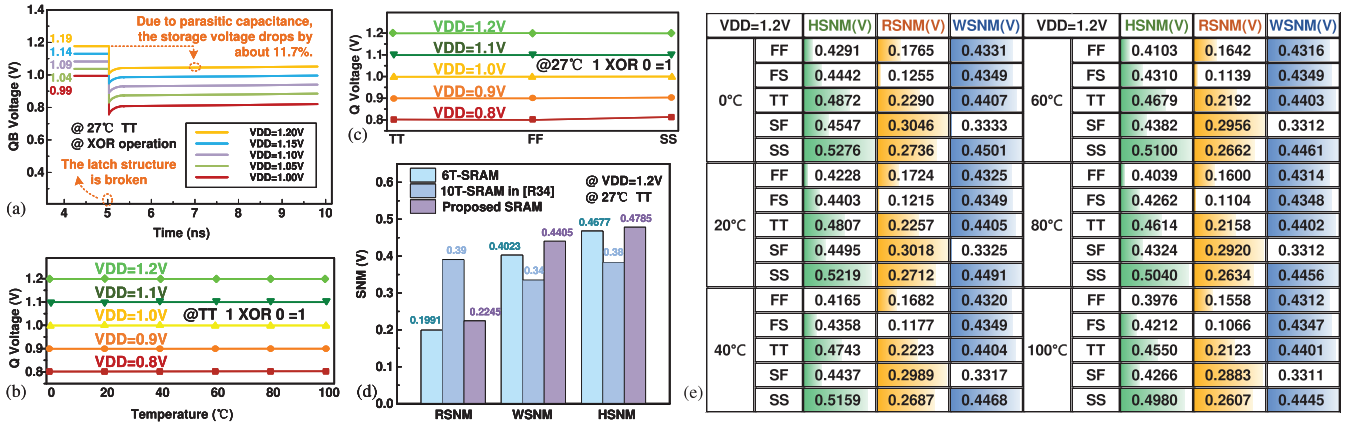


Fig. 5. (a) OR operation procedure, (b) OR operation truth and mapping table, (c) XOR operation procedure, (d) XOR operation truth and mapping table, (e) IMP operation procedure, and (f) IMP operation truth and mapping table.



VDD=1.2V		HSNM(V)	RSNM(V)	WSNM(V)	VDD=1.2V		HSNM(V)	RSNM(V)	WSNM(V)
0°C	FF	0.4291	0.1765	0.4331	60°C	FF	0.4103	0.1642	0.4316
	FS	0.4442	0.1255	0.4349		FS	0.4310	0.1139	0.4349
	TT	0.4872	0.2290	0.4407		TT	0.4679	0.2192	0.4403
	SF	0.4547	0.3046	0.3333		SF	0.4382	0.2956	0.3312
	SS	0.5276	0.2736	0.4501		SS	0.5100	0.2662	0.4461
20°C	FF	0.4228	0.1724	0.4325	80°C	FF	0.4039	0.1600	0.4314
	FS	0.4403	0.1215	0.4349		FS	0.4262	0.1104	0.4348
	TT	0.4807	0.2257	0.4405		TT	0.4614	0.2158	0.4402
	SF	0.4495	0.3018	0.3325		SF	0.4324	0.2920	0.3312
	SS	0.5219	0.2712	0.4491		SS	0.5040	0.2634	0.4456
40°C	FF	0.4165	0.1682	0.4320	100°C	FF	0.3976	0.1558	0.4312
	FS	0.4358	0.1177	0.4349		FS	0.4212	0.1066	0.4347
	TT	0.4743	0.2223	0.4404		TT	0.4550	0.2123	0.4401
	SF	0.4437	0.2989	0.3317		SF	0.4266	0.2883	0.3311
	SS	0.5159	0.2687	0.4468		SS	0.4980	0.2607	0.4445

Fig. 6. (a) Simulation of voltage maintenance stability at the storage node. (b) Voltage stability under varying temperatures and supply voltages. (c) Voltage stability across different process corners and supply voltages. (d) RSNM/WSNM/HSNM comparison among the traditional 6T-SRAM, the 10T-SRAM in [34], and the proposed structure. (e) Simulation results of RSNM/WSNM/HSNM for the proposed 10T cell at different temperatures and process corners.

the self-recycling 10T-SRAM cell, followed by a performance comparison with traditional SRAM structures.

1) Stability of Storage Nodes During CIM Mode Operation: As described in the previous text, this structure performs the logical operation function in four sequential steps: the left-half calculation stage, the holding stage, the right-half calculation stage, and the final latching stage. During the holding stage, the cross-coupled latch structure is disconnected, necessitating an evaluation of the extent to which the stored data are affected.

As shown in Fig. 6(a), under the TT process corner and a temperature of 27 °C, the voltage level of the storage node QB at varying power supply voltages is illustrated. The simulated scenario represents the XOR operation, where the input IN1 is "1," IN2 is "0," and Q is "0," corresponding to 0 XOR 0 = 0. During the second step of the operation, the storage node QB is required to maintain a high level. However, due to the disconnection of the cross-coupled latch structure, the node is influenced by parasitic capacitance, resulting in a voltage drop of only 11.7%. Compared with 20.5% of that in the CIM structures [13], this design shows a significant improvement. Furthermore, since the holding time during this process is relatively short in actual simulation, and the subsequent calculation is performed before the node can flip, the accuracy of the calculation results remains unaffected.

The stability of node voltages during computation is critically influenced by PVT variations. We selected the 1 XOR 0 = 1 operation as a representative test case and performed simulations across varying supply voltages (0.8–1.2 V), temperature ranges (0 °C–100 °C), and process corners (TT/FF/SS). The simulation results are presented in Fig. 6(b) and (c), which clearly demonstrates that the voltage values at node Q after the latching stage finished

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converge to within 2% of VDD (0.98–1.02 VDD) under all tested conditions, ensuring the accuracy of the computational results.

2) Read, Write, and Hold Noise Margin Measurement: Static noise margin (SNM) can evaluate the anti-interference ability of SRAM cells and can be divided into three types: read SNM (RSNM), write SNM (WSNM), and hold SNM (HSNM),.

For traditional 6T-SRAM, there is a problem of read disturbance during the read operation, and it is easily affected by some external noise and has poor anti-interference ability. Its noise margin SNM is typically related to several factors, including the power supply voltage, the threshold voltage of the transistor, the ratio of the width-to-length ratio of the driving transistor to the access transistor, and the ratio of the width-to-length ratio of the load transistor to the access transistor.

Fig. 6(d) presents the RSNM, WSNM, and HSNM of the self-recycling 10T-SRAM cell proposed in this article, the conventional 6T-SRAM cell, and the 10T-SRAM in [34]. During the data read process, the storage cell may be susceptible to external noise. If the noise voltage exceeds the flipping voltage of the internal inverter, it may compromise the accuracy of the read operation. From the data, it can be observed that the RSNM of the traditional 6T-SRAM structure is only 0.1991 V, and that of the 10T-SRAM in [34] is 0.39 V, whereas the RSNM of the proposed self-recycling 10T-SRAM structure is 0.2245 V. This indicates that although the RSNM of the proposed 10T structure is lower than that of the 10T structure in [34], it still demonstrates a 0.0254-V improvement over conventional 6T-SRAM.

In addition, write static noise margin and hold static noise margin of the self-recycling 10T-SRAM cell proposed in this article are both increased compared to those of the traditional 6T-SRAM cell and the 10T-SRAM reported in [34]. The WSNM increases by 38.2 mV from 0.4023 V of the traditional 6T-SRAM structure and 100.5 mV from 0.34 V of the 10T-SRAM reported in [34] to 0.4405 V of the proposed 10T-SRAM. Furthermore, the HSNM of the traditional 6T-SRAM structure is 0.4677 V, and that of the 10T-SRAM reported in [34] is 0.38 V, while the HSNM of the self-recycling 10T-SRAM structure proposed in this article is 0.4785 V. Since the antinoise ability of the cell is proportional to the noise margin value, in conclusion, the self-recycling 10T-SRAM structure has stronger antinoise ability.

Fig. 6(e) shows the HSNM, RSNM, and WSNM at different temperatures and process corners when the power supply voltage VDD is 1.2 V. It can be observed that as the temperature increases, the noise margin value decreases progressively under the same power supply voltage and process corner.

At the same power supply voltage and temperature, the TT process corner exhibits moderate performance. The FS and SF process corners represent the extremes, with one being the best and the other the worst. This behavior occurs because, in the FS process corner, the nMOS transistors are fast, while the pMOS transistors are slow. In the SF process corner, the pMOS transistors are fast, and the nMOS transistors are slow.

B. Simulation of the Robustness of Logical Operation Result

The robustness of the proposed self-recycling 10T-SRAM cell is analyzed using ADE XL and 3000-trail Monte Carlo (MC) simulations for four logical operations: AND, OR, XOR, and implication (IMP).

1) AND *Operation [See Fig. 7(a)]:* At a power supply voltage VDD =1.2 V and the TT process corner at 27 °C, the MC analysis shows an average output voltage of 1.198 V with a standard deviation of 0.84 mV. All 3000 results are correct, confirming the accuracy and robustness of the AND logical operation.
2) OR *Operation [See Fig. 7(b)]:* Under the same conditions, the average output voltage for 3000 simulations is 1.198 V, with a smaller standard deviation of 0.68 mV. The node voltages are tightly concentrated, and no errors are observed, ensuring reliable performance for the OR logical operation.
3) *IMP Operation [See Fig. 7(c)]:* The MC simulation results for the IMP logical operation yield an average output voltage of 3.8 mV with a standard deviation of 1.15 mV. The node voltages follow a normal distribution, and all results are correctly flipped without errors, indicating robust and accurate operation.
4) XOR *Operation [See Fig. 7(d)]:* At VDD =1.2 V, the XOR operation demonstrates an average output voltage of 4.1 mV with a standard deviation of 1.17 mV. The node voltages conform to a normal distribution, and all results are correctly flipped, confirming the robustness and accuracy of the XOR logical operation.

C. Simulation of the Computational Performance of Full Array Activation

The evaluation of the logical operation of the proposed self-recycling 10T-SRAM architecture primarily focuses on energy consumption and speed analysis across four types of logical operations. The speed analysis encompasses simulations of calculation delay and operating frequency.

1) Simulation of Logical Operation Energy Consumption: The CIM macro proposed in this article supports four full-array in-memory logical operations: AND, OR, XOR, and IMP. Unlike conventional SRAM, where power consumption is dominated by global bitline precharging, the proposed architecture performs logic operations and writes results within the cell, eliminating energy-intensive bitline operations. Furthermore, conventional recycling often involves activating a large number of cells unnecessarily, leading to inefficient power use. In contrast, the proposed design selectively activates only the required cells for each operation, further enhancing energy efficiency.

In addition, the proposed CIM macro employs parallel array computation to execute multiple logical operations simultaneously, improving throughput. When normalized, the power per operation equals the total array power divided by the number of cells, reducing average energy consumption compared to sequential SRAM access. Although additional control logic introduces some power overhead, the architecture minimizes this through an optimized design. Specifically, in the

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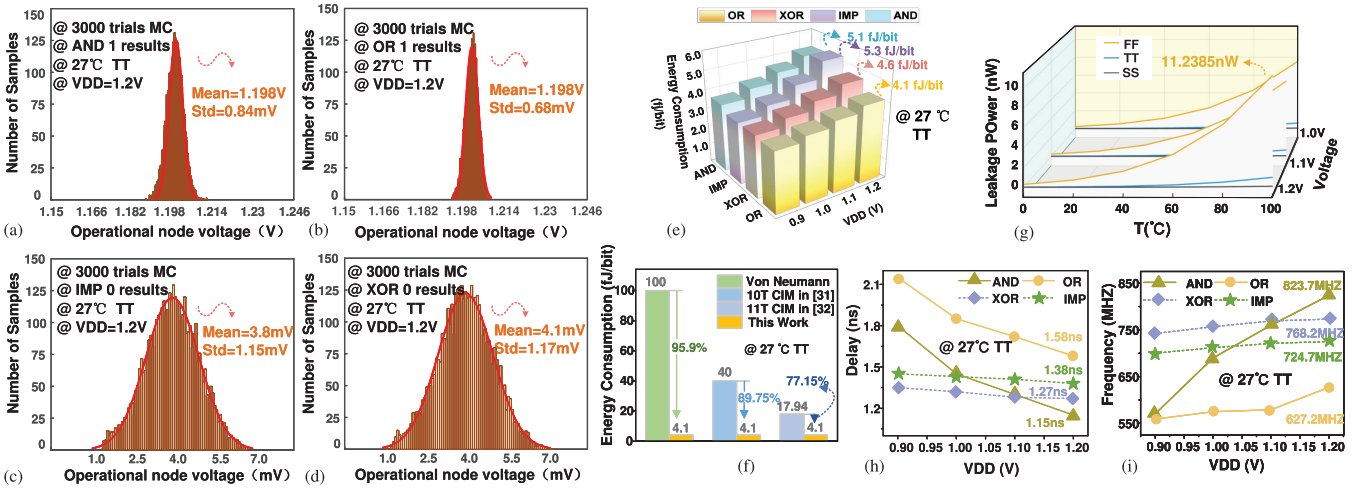


Fig. 7. (a) 3000-trial MC simulation results for the AND operation. (b) 3000-trial MC simulation results for the OR operation. (c) 3000-trial MC simulation results for the IMP operation. (d) 3000-trial MC simulation results for the XOR operation. (e) Energy consumption comparison of four logical operations. (f) Energy consumption comparison between the proposed architecture and non-self-recycling architectures. (g) Leakage power consumption at different process corners, voltages, and temperatures. (h) Delay comparison of logical operations. (i) Frequency comparison of logical operations.

proposed four-step computation per operation, control lines CL and CN require only four switching transitions each, while lines RCL and RCN require just two transitions. This represents a substantial improvement over the baseline design, which demands eight transitions per control line, achieving a 50%–75% reduction in control-line switching activity and significantly lowering the power consumption of the control subsystem.

Fig. 7(e) illustrates the energy consumption simulation of these four logical operations under full array activation, at a temperature of 27 °C and a process corner of TT, across different power supply voltages. At 1.2 V, the energy consumption for the OR, XOR, AND, and IMP operations is 4.1, 4.6, 5.1, and 5.3 fJ/bit, respectively.

It can be known from Fig. 7(e) that the energy consumption of the full array activation operation decreases as the power supply voltage drops. This is because the dynamic energy consumption is directly proportional to the voltage. The smaller the voltage is, the lower the energy consumption will be.

Fig. 7(f) shows the energy consumption comparison between the proposed self-recycling architecture and the non-self-recycling architectures, including the von·Neumann architecture, the 10T-SRAM CIM architecture [31], and the 11T-SRAM CIM architecture [32]. The proposed self-recycling CIM macro achieves significant energy savings, demonstrating 95.9% reduction compared to the von Neumann architecture (100 fJ/bit), 89.75% reduction relative to the 10T-SRAM design (40 fJ/bit) [31], and 77.15% improvement over the 11T-SRAM implementation (17.94 fJ/bit) [32]. This demonstrates its remarkable high energy-efficiency characteristics.

Furthermore, static leakage power dissipation in SRAM is a critical metric requiring focused attention, which is caused by subthreshold leakage, junction leakage, and gate leakage. Among these, subthreshold leakage is the predominant factor in the circuit. Fig. 7(g) shows the static power dissipation of the suggested design at various cell supply voltages, temperatures, and process corners. It is evident that the static leakage power consumption exhibits a monotonic increase with higher supply voltages or elevated temperatures while also demonstrating pronounced sensitivity to process corner variations illustrated in Fig. 7(g). It reveals that the proposed design generates up to 11-nW leakage power under the FF process corner. However, this reduces to 0.022–1.005 nW in the TT process corners and further decreases to 0.115 nW (maximum) in SS corners, which are lower than the leakage power above 1.2 nW reported for the 10T-SRAM structure in [34] whose VDD =1.0 V under TT process corner.

2) Simulation of Logical Operation Delay and Frequency: Fig. 7(h) is the delay analysis diagram of AND, OR, XOR, and IMP logical operations. Fig. 7(i) is the frequency analysis diagram of the four logical operations. Through these two figures, we can obtain that at a temperature of 27 °C and under the TT corner, when the power supply voltage is 1.2 V, the delays of the AND, OR, XOR, and IMP operations are 1.15, 1.58, 1.27, and 1.38 ns, and the frequencies of the AND, OR, XOR, and IMP operations are 823.7, 627.2, 768.2, and 724.7 MHz.

As can be seen from the two figures, there is such a trend: as the power supply voltage decreases, the frequency decreases and the delay increases. This is because, as the power supply voltage decreases, the response speed of the transistor will become slower and slower, the delay will correspondingly increase, and the overall calculation time will also become longer. Frequency is inversely proportional to the period. Therefore, the frequency will also decrease correspondingly.

D. Performance Comparison of the Architectures for Application

In the era of big data, data encryption has become increasingly critical [13], [24], [30]. Encryption and decryption algorithms are essential components of modern computer networks. Typically, multiple rounds of encryption are applied

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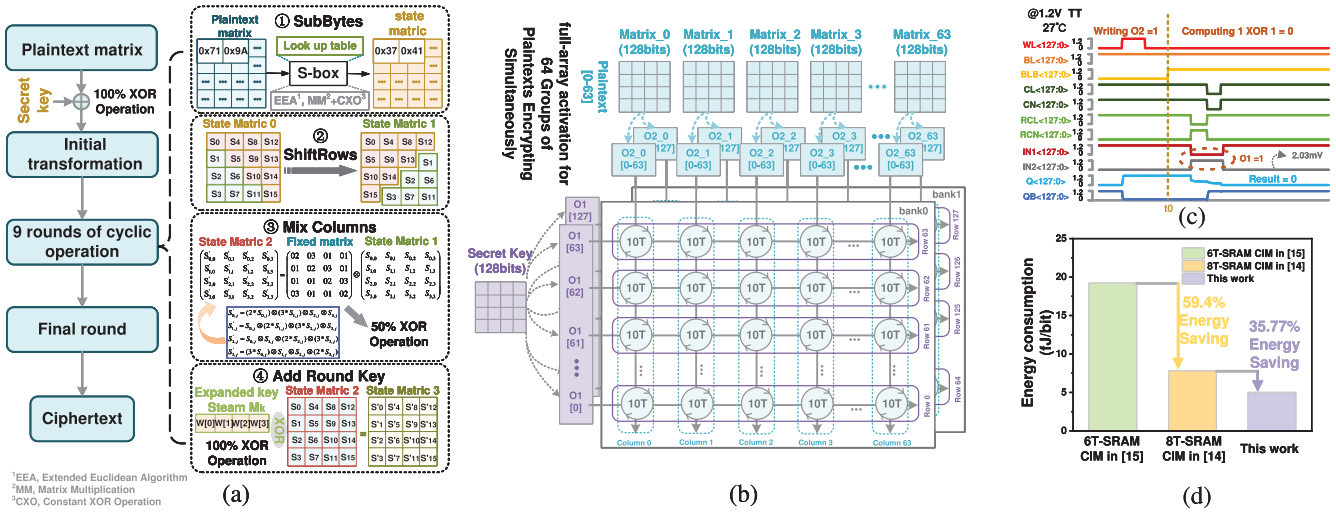


Fig. 8. (a) Flowchart of AES algorithms. (b) Mapping strategy of the encryption process. (c) Timing diagrams of XOR operation for AES algorithms. (d) Energy consumption comparison of the architectures for AES.

to plaintext to produce ciphertext, ensuring both the security and integrity of the data.

The self-recycling 10T-SRAM CIM macro proposed in this article can be applied to the AES. Fig. 8(a) shows the flowchart of the AES algorithm. Prior to the initial round, the plaintext is converted into a 128-bit encryption block and multiround processing based on key length: ten rounds for 128-bit keys, 12 rounds for 192-bit keys, and 14 rounds for 256-bit keys. When configured with the key matrix, the AES encryption process initiates with an initial transformation phase composed entirely of XOR operations. This is followed by nine rounds of cyclic encryption and a final round. Each of the nine rounds includes the following four steps: subbytes that bytewise substitution via S-box, shift rows for diffusion, mix columns involving 50% XOR operations between the data state and a fixed matrix, and add round key with 100% XOR operation. Thus, the AES encryption algorithm heavily relies on substitution-and-permutation operations employing bitwise Boolean operations such as XOR logic.

We selected initial transformation (XOR operation between plaintext and fixed key matrix) as an example showing the mapping strategy of the encryption process illustrated in Fig. 8(b). To support the multiround encryption requirements of AES, the proposed 64 × 64 CIM array adopts a scalable multibank architecture, where the number of banks can be dynamically configured to match different security levels: two banks for ten-round AES-128, three banks for 12-round AES-192, and so on. This demonstrates the excellent scalability of the proposed design. The operational workflow begins with loading 64 groups of 128-bit plaintext matrices (Plaintext [0–63]) into the CIM array as operand2 (O2) in columnwise. Among them, the lower 64 bits (O2[0–63]) are stored in Bank 0 (LSB-aligned 64 × 64-bit subarray), and the upper 64 bits (O2[64–127]) are stored in Bank 1 (MSB-aligned 64 × 64-bit subarray). Then, the 128-bit key matrix is written as operand1 (O1) in rowwise, and the XOR logical operation is performed on the full array leading to 64 groups of plaintexts being encrypted simultaneously. Finally, the encrypted text is obtained by reading columnwise.

The scalability of the proposed architecture introduces inherent challenges associated with interconnect resistance escalation, signal integrity degradation, and power delivery network (PDN) limitations, which may adversely affect computational performance and energy efficiency. Fortunately, extensive simulations validate the functional robustness of the scaled design. As demonstrated in Fig. 8(c), operand2 is accurately programmed into storage node Q at t0, followed by a correct XOR logic computation ("1 XOR 1 = 0"). Notably, the Q-node voltage stabilizes at <10 mV (near-rail-to-rail swing), confirming preserved computational fidelity despite scaled parasitics.

However, architectural expansion incurs an inevitable energy overhead. As shown in Fig. 8(d), the 8-kb 64 × 128 CIM macro, which is configured for XOR operations of AES-algorithmic, reveals an energy consumption of 5.01 fJ/bit, representing an 8.91% increase compared to the baseline 64 × 64 CIM macro (4.6 fJ/bit). This achieves a 35.77% energy saving over the 8T-SRAM CIM architecture (7.8 fJ/bit) [14], which represents a 59.4% reduction compared to the 6T-SRAM CIM architecture [15] (19.21 fJ/bit). This marginal energy penalty remains within acceptable thresholds for memory-intensive cryptographic workloads.

Afterward, Table I presents a comparison between the proposed 10T-SRAM self-recycling architecture and other state-of-the-art in-memory computing architectures. The proposed architecture features automatic write-back of operation results and supports multiple logical operations directly within SRAM.

The maximum operation frequency of this architecture is 823.7 MHz, which is higher than 313.7 MHz (2.63×) in [12]. The lowest calculation and write-back energy consumption of this architecture is 4.1 fJ/bit. Compared with the calculation

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TABLE I
COMPARISON TABLE WITH RELATED RESEARCH PERFORMANCE

	JSSC2021 [11]	SSCL2023 [12]	ISSCC2022 [15]	VLSI2022 [14]	JSSC2022 [13]	ISCAS2024 [31]	TCAS II 2024 [32]	This work
Technology (nm)	28	55	28	28	28	40	65	**55**
Cell type	10T	10T	6T	6T/8T	8T	10T	11T	**10T**
Array size	4Kb	4Kb	32Kb	16Kb	16Kb	4Kb	4Kb	**4Kb**
Supply voltage (V)	0.7 - 0.9	1.2	0.8	0.5 - 0.9	0.65	1.0	1.2	**1.2**
Frequency	300 MHz (0.9 V)	313.7 MHZ	333 MHz	600 MHz (0.6 V)	29 MHz (0.6 V) 113 MHz (0.75 V)	629MHZ	595MHZ	**823.7 MHZ**
Logic energy (fJ/bit)	15 (0.9V) 10.5 (0.7 V)	2.75 (computing-restoring [4])	19.21-35.55	5.7 (0.5 V) 7.8 (0.6 V)	3.7 (0.66V)	40(1.0V)	17.94(1.2V)	**4.1 (computing-restoring)**
Energy efficency (TOPS/W)	66.67 (0.9 V) 95.2 (0.7 V)	364.2 (computing-restoring)	27.38 (8b/8b)	175.4 (0.5 V) 128.2 (0.6 V)	270.5 (0.66V)	25	55.74	**246.9 (computing-restoring)**
Throughput [1] (GOPS/Kb)	9.6	321	11.8	39	115.7 (0.75V)	/	/	**843** [3]
Self-recycling	NO	YES	NO	NO	YES	NO	NO	**YES**
Type of supported functions	SRAM/CAM [2]/ Logic/Matrix transpose	SRAM/Logic/ Self-recycling	SRAM/Logic (AND/OR/XOR)	SRAM/Logic/ CAM	SRAM/In-situ/ Logic/ Copy operation/ Matrix transpose	CAM/ Logic/ Arithmetic/ Matrix transpose	SRAM/Logic/ CAM/ Matrix transpose	**SRAM/Logic (AND/OR/IMP/XOR)/ Self-recycling**

[1] Scale with operands bitwise length
[2] CAM, content-addressable memory
[3] Calculated as 64×64 Op/cycle × 1 Op × 823.7 M cycles / 4Kb
[4] Calculated as 64×64 Op/cycle × 1 Op / (VDD × $I_{Average}$); $I_{Average}$: Mean current in computing-restoring cycle

mode in [31], the energy consumption is reduced by 89.75%. The reason why it is higher than 3.7 fJ/bit in [13] is that VDDs are different. The energy efficiency of this architecture is 246.9 TOPS/W. The throughput of this architecture is 843 GOPS/kb, which is 2.64× higher than that in [12].

IV. CONCLUSION

In this article, we propose a self-recycling 10T-SRAM-based CIM architecture capable of performing logical operations and full-array activation without additional memory while restoring CIM results directly in the bit-cell. Full-array Boolean logical operations, including AND/NAND, OR/NOR, XOR/XNOR, and IMP, are achieved via two operands orthogonal writing, enabling activation across the entire array and significantly enhancing throughput to 843 GOPS/kb. System architecture simulations were conducted using 55-nm CMOS technology, where HSNM, WSNM, and RSNM values of 478.5, 440.5, and 224.5 mV were observed at 1.2 V and 27°C, respectively. The energy consumption for full-array Boolean logic and self-recycling operations was as low as 4.1 fJ/bit, achieving energy efficiency of 246.9 TOPS/W at a 1.2-V supply voltage. When applied to AES algorithms, the proposed circuits reduced energy consumption by at least 35.77% compared to the digital CIM architecture that is not self-recycling.

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